

Eduardo José Duarte Peña

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Machine Learning Engineer · AI Engineer · Cloud (Azure & AWS)

Professional Summary

Machine Learning Engineer with a Mechatronics background, specialized in designing **end-to-end ML and AI systems**—from data acquisition and feature engineering to model training, optimization, and production deployment on **Microsoft Azure and AWS**. Hands-on experience building real-time inference pipelines, distributed edge-ML systems, and applied AI in signal-processing domains. Combines rigorous mathematical foundations with production-grade Python, FastAPI, PyTorch, and MLOps practices to deliver reliable systems in multinational engineering environments.

Education

Universidad Santo Tomás

B.Sc. in Mechatronics Engineering

Bucaramanga, Colombia

2019–2025

Google – Coursera

Data Analytics Professional Certificate

2023

Professional Experience

Space-Eyes

Machine Learning Engineer

Remote

Oct 2025 – Present

- Trained and optimized **deep-learning models** on **Microsoft Azure** (Azure ML, AKS) and **AWS** (EC2, S3), enabling reproducible training workflows and scalable inference across cloud providers.
- Architected an **end-to-end ML pipeline**—data acquisition, feature engineering, model training, and edge deployment—for real-time signal classification on resource-constrained devices.
- Engineered an **ML-driven distributed detection & localization system** on edge hardware (Raspberry Pi 5), integrating model inference with custom data-acquisition stacks.
- Defined the **mathematical foundations for ML-based geolocation algorithms** (TDOA, GDOP, DOA estimation), producing internal reference documentation adopted as team standard.
- Designed a **geometric coverage-optimization algorithm** (multi-ring node placement) guaranteeing ≥ 2 -node overlap for reliable inference results.
- Deployed a **resilient MQTT distributed architecture** sustaining stable telemetry and data transmission over distances up to **5 km** for real-time model inputs.
- Built a **scalable FastAPI backend** exposing model predictions and data through REST APIs, and a **React frontend** for real-time visualization, system health, and geospatial inference output.
- Led the **transition from MVP to Tier-1 production architecture**, defining system tiers, eliminating bottlenecks, and improving scalability, modularity, and reliability of the ML platform.
- Managed version control and structured code-review workflows on GitHub.

ELVA AI

Machine Learning Engineer

Remote

Aug 2025 – Sept 2025

- Trained a **multi-class deep-learning classifier** (Wi-Fi, LTE, DVB-T) from raw input samples, optimized for resource-constrained edge devices.
- Automated **city-level model performance reports** including confusion matrices and interactive visualizations, eliminating hours of manual reporting.
- Hardened cloud deployment by implementing **Azure AD authentication on AKS**, with complementary ML workloads on **AWS** (S3 storage, IAM-based access control).
- Reduced cloud storage costs by **20%** through disk-tier optimization and lifecycle/retention policies across Azure and AWS.

Tayronama SAS

Automation Developer & Data Analyst

Mar 2025 – Jul 2025

- Developed a **Python data-acquisition application** for livestock traceability with real-time integration via serial communication.
- Engineered ETL workflows in Pandas and built **Power BI dashboards** surfacing operational KPIs to stakeholders.
- Automated client report delivery through the **WhatsApp Business API**, eliminating manual dispatch and improving communication SLAs.

Merck Sharp & Dohme (MSD)

Innovation & Technology Intern – Data Analytics

Jul 2024 – Feb 2025

- Saved over **30 hours per month** by re-engineering training processes through data-driven analysis.
- Built a **KPI monitoring system in Power BI** ensuring continuous performance visibility for leadership.
- Delivered automation solutions with **Power Apps and Power Automate**, increasing operational efficiency across multiple workflows.
- Processed large pharmaceutical datasets using **Python, SQL, and ML libraries** to support evidence-based decisions.

Universidad Santo Tomás

Junior Data Scientist / Researcher

Jun 2023 – Nov 2023

- Built **ML models** predicting public-transportation routing patterns in Bucaramanga.
- Delivered an interactive **Power BI dashboard** surfacing urban mobility insights for civic stakeholders.
- **Co-author**, *Advances in solutions on public transportation problems with AI* (peer-reviewed publication).

Technical Skills

Machine Learning & AI: PyTorch, scikit-learn, Deep Learning, Model Training & Optimization, Feature Engineering, Signal Classification, MLOps

Programming: Python, SQL, JavaScript / React, FastAPI

Cloud & Infrastructure: Microsoft **Azure** (Azure ML, AKS, Azure AD), **AWS** (EC2, S3, IAM), Edge Computing, Docker, Cloud Deployment

Signal Processing & Hardware: IQ Signal Processing, Spectrogram Generation, SDR (bladeRF), Raspberry Pi, GPSDO, TDOA / GDOP / DOA Estimation, GCC-PHAT, MUSIC / ESPRIT

Distributed Systems: MQTT, Real-Time Telemetry, Edge-Cloud Pipelines, Long-Range Communication

Data & Analytics: Pandas, NumPy, Power BI, Power Apps, Power Automate, Data Visualization

Tools & Methods: Git / GitHub, CRISP-DM, Agile, Process Automation

Awards

2023: 1st Place – CienTloN, Research & Innovation Category

2023: 3rd Place – Huawei ICT Competition, Cloud Track

Certifications

2024: Process Improvement – Harvard ManageMentor

2023: Data Science & AI – MinTIC

2023: Machine Learning Fundamentals – DataCamp

2023: Implementing AI Solutions in Business – DataCamp

2023: Databases and SQL for Data Science – Coursera

2023: Deep Learning in Python – DataCamp

2022: Python for Data Science – Coursera

Languages

Spanish: Native

English: B2 – Professional Working Proficiency

References

Sebastián Murcia – General Manager, Tayronama SAS – +57 321 6680277

Iván Lis – Safety Data Manager, MSD – +57 310 7735259

Carlos Pimentel – Research Professor, Universidad Santo Tomás – +57 316 2373049